

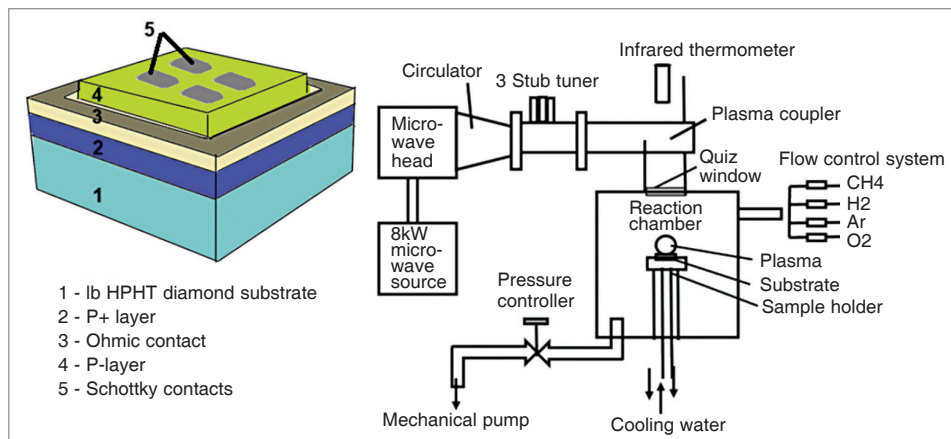
Makoto Kasu, began investigating diamond semiconductors. Saga University's pioneering research has produced promising results that could pave the way for the use of diamond semiconductor technology in such applications.

Over 190 hours of continuous use, Kasu's power circuit demonstrated stability without deterioration, switching in less than 10 nanoseconds. The properties of diamond as a semiconductor make it a potential material for future electronic components. This expands the range of operating conditions for CPUs and removes the need for direct cooling.

The researchers constructed a transistor using two 'phosphorus-doped diamond epilayers.' Phosphorus doping—basically, the addition of the element to the layers—is required to increase conductivity. In a traditional chip, the silicon-based layer would be replaced with this n-channel layer, which transports free electrons. Through this process, enough electrons flow to bridge two ends of a gate, called 'the source' and 'the drain,' which closes the circuit to indicate a 1 instead of a 0.

The team heavily doped the second, positive layer and lightly doped the first, negative layer with phosphorus. The use of the second diamond layer significantly reduced source and drain contact resistance. The researchers then added aluminium trioxide, 30 nanometres thick, as an insulator after annealed titanium 'source' and 'drain' connections were formed on the top, heavily doped layer. The result was the first functional n-channel MOSFET transistor made with diamond.

The transistor was subjected to various tests to evaluate its conductivity performance. The researchers



Schematic diagram of MPCVD system

reported that the n-type diamond MOSFETs demonstrated high field-effect mobility of around $150\text{cm}^2/\text{V}\cdot\text{sec}$ at 573K—the highest among all n-channel MOSFETs based on wide-bandgap semiconductors—indicating strong conductivity and stability at high temperatures.

Related companies are gaining more attention due to the potential for diamond semiconductors to accelerate commercialisation. Notable pioneers include:

Adamant Namiki Precision Jewel Co, Ltd (Ad-Na). A Japanese company specialising in producing high-quality diamond wafers essential for semiconductor device fabrication

Saga University. This Japanese university has extensively studied diamond semiconductors and developed cutting-edge devices with record-breaking output power

Element Six. A leader in the creation and synthesis of synthetic diamonds, particularly for electronic devices

Cree. A semiconductor manufacturer that has investigated diamond substrates for high-power electronics

Power semiconductors are the engine driving the growth of more electric economies, managing and modifying electricity flows. Their impact is essential for system

power, compactness, and efficiency. Thus, they are indispensable for widespread electrification and the decarbonisation of entire economic sectors. Emerging 'wide bandgap' diamond semiconductors are considered the ultimate power semiconductor in Japan, Europe, and the US, making them the preferred material for a new generation of power management components.

Diamond semiconductor by MPCVD

Microwave plasma chemical vapour deposition (MPCVD) is a non-polar discharge diamond-coating technique that uses microwave plasma to control the deposition process of thin films, allowing for precise control over factors such as temperature, pressure, and the choice of precursor gas. This ultimately forms a diamond film on the substrate. The steps involved in creating a diamond semiconductor are as follows:

Gas mixture. Methane and hydrogen are mixed and introduced into the chamber.

Microwave excitation. Methane is broken down by the activation energy from microwave power, which creates plasma that causes a glow discharge.

Carbon deposition. The carbon atoms produced in this process are